the model, we have a category of functions, and we still have to learn the parameters that tell us which specific function in that category is the best for our dataset.

For example, if we choose a generic line, then we still have to decide the values for the parameters `m` and `b`: it could be `y = x + 1`, or just as well `y = -0.5*x +42`.

The way we choose those is through **training**: nothing else than applying gradient descent.

In linear regression, in fact, we define a cost function (usually referred to as "**loss function**" in machine learning) that measures the distance between the value predicted by the model for the dependent variable associated to each point in the dataset[16], and the actual value from the data.

This function is generally the sum of least square errors, or some variant of it; as shown in figure 16.17 and 16.19, we minimize the squared distance, along the y axis, between each point and the model line, and this gives us a convex, bowl shaped function with a global minimum: as we have seen, that's pure gold for gradient descent!